

Linear Algebra – MAT 2610

Section 3.1 (Introduction to Determinants)

Dr. Jay Adamsson

jay@upei.ca

jadamsson@upei.ca



Chapter 2, Theorem 4

If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ with $ad - bc \neq 0$ then

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \leftarrow$$

The value $ad - bc$ is called the **determinant** of A and is denoted $\det(A)$

Note that the matrix A has an inverse if and only if $\det(A) \neq 0$



Determinant of 2X2 Matrix

$$\det \begin{pmatrix} 2 & -1 \\ 3 & -4 \end{pmatrix} = (2)(-4) - (-1)(3) = \underline{(-8) - (-3)} = \underline{-5}$$

Use the determinant to determine for what values of s is $\begin{bmatrix} 8-s & -2 \\ 4 & 2-s \end{bmatrix}$ invertible

$$\det \begin{bmatrix} 8-s & -2 \\ 4 & 2-s \end{bmatrix} = (8-s)(2-s) - (-2)(4) = [16 - 10s + s^2] - (-8)$$

$$= s^2 - 10s + 24 = (s-6)(s-4)$$

So the matrix has no inverse if $(s-6)(s-4) = 0$
or when $s=6$ or $s=4$

The matrix has an inverse if $s \neq 6$ and $s \neq 4$



Cofactor Expansion

Definition: For any $n \times n$ matrix A , the submatrix formed by deleting the i -th row and the j -th column is termed $A_{i,j}$.

Definition: For any $n \times n$ matrix A , the (i, j) -*cofactor* of A is denoted $C_{i,j}$ and is calculated as $C_{i,j} = (-1)^{i+j} \det(A_{i,j})$.



Example

If $A = \begin{bmatrix} 2 & 1 & 2 \\ 4 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$, then $A_{2,3}$ is created by removing the 2nd row and the 3rd column,
giving $A_{2,3} = \begin{bmatrix} 2 & 1 \\ 2 & 2 \end{bmatrix}$ and $C_{2,3} = (-1)^{2+3} \det(A_{2,3}) = (-1)^5 (2 * 2 - 1 * 2) = -2$



Definition

For any $n \times n$ matrix A , the determinant is given by cofactor expansion along the first row:

$$\det(A) = a_{1,1}C_{1,1} + a_{1,2}C_{1,2} + \cdots + a_{1,n}C_{1,n}$$

Handwritten red annotations: Under $a_{1,1}$ is a wavy line and a bracket. Under $a_{1,2}$ is a wavy line and a bracket. Under $a_{1,n}$ are two brackets.



Exercise

If $A = \begin{bmatrix} 2 & 1 & 2 \\ 4 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$, calculate $\det(A)$

$$\det = a_{11} C_{11} + a_{12} C_{12} + a_{13} C_{13}$$

$$a_{11} = 2$$

$$a_{12} = 1$$

$$a_{13} = 2$$

$$C_{11} = (-1)^{1+1} \det \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix} = (-1)^2 [(2)(3) - (1)(2)]$$
$$= (1)(6 - 2) = 4$$

$$C_{12} = (-1)^{1+2} \det \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} = (-1)^3 [(4)(3) - (1)(2)]$$
$$= (-1)(12 - 2) = -10$$

$$C_{13} = (-1)^{1+3} \det \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix} = (-1)^4 [(4)(2) - (2)(2)]$$
$$= (1)[8 - 4] = 4$$



$$A = \begin{bmatrix} 2 & 1 & 2 \\ 4 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$$

$$\begin{aligned} \det(A) &= a_{11} C_{11} + a_{12} C_{12} + a_{13} C_{13} \\ &= (2)(4) + (1)(-10) + (2)(4) \\ &= 8 - 10 + 8 = \underline{\underline{6}} \end{aligned}$$



Theorem 1

The determinant of any $n \times n$ matrix A can be computed by a cofactor expansion across any row or down any column. The expansion across the i^{th} row using the cofactors is

$$\det(A) = \underline{a_{i1}} \underline{C_{i1}} + \cdots + \underline{a_{in}} \underline{C_{in}} \quad]$$

The cofactor expansion down the j^{th} column is

$$\det(A) = \underline{a_{1j}} \underline{C_{1j}} + \cdots + \underline{a_{nj}} \underline{C_{nj}} \quad]$$



Example

If $A = \begin{bmatrix} 2 & 1 & 2 \\ 4 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$, calculate its determinant using cofactor expansion along the second column.

$$C_{12} = (-1)^{1+2} \det \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} = (-1)^3 (4 \cdot 3 - 1 \cdot 2) = (-1)(10) = -10$$

$$C_{22} = (-1)^{2+2} \det \begin{bmatrix} 2 & 2 \\ 2 & 3 \end{bmatrix} = (-1)^4 (2 \cdot 3 - 2 \cdot 2) = (1)(2) = 2$$

$$C_{32} = (-1)^{3+2} \det \begin{bmatrix} 2 & 2 \\ 4 & 1 \end{bmatrix} = (-1)^5 (2 \cdot 1 - 2 \cdot 4) = (-1)(-6) = 6$$

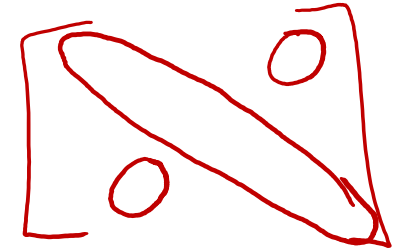
$$\begin{aligned} \det(A) &= a_{12} C_{12} + a_{22} C_{22} + a_{32} C_{32} \\ &= (1)(-10) + (2)(2) + (2)(6) \\ &= -10 + 4 + 12 = 6 \end{aligned}$$



Definition: Let A be an $n \times n$ matrix. The **main diagonal** of A consists of the entries $a_{11}, a_{22}, \dots, a_{nn}$. A is a **diagonal matrix** if all entries off its main diagonal are 0.

$$\begin{bmatrix} 3 & 0 \\ 0 & -4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$



Definition: Let A be an $n \times n$ matrix. A is **upper triangular** if all of its entries to the left and below the main diagonal are 0. A is **lower triangular** if all of its entries to the right and above the main diagonal are 0.

Upper triangular:

$$\begin{bmatrix} 1 & 2 & 4 \\ 0 & -2 & -1 \\ 0 & 0 & 4 \end{bmatrix}$$

Lower triangular:

$$\begin{bmatrix} 1 & 0 & 0 \\ -7 & -2 & 0 \\ 3.5 & 0 & 2 \end{bmatrix} = 0$$

Upper and Lower triangular:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$



Example: Find $\det(A)$ where $A =$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 4 & -2 & 0 & 0 \\ 1 & -2 & 3 & 0 \\ -1 & -2 & 500 & 4 \end{bmatrix}$$

$$\begin{aligned} \det(A) &= a_{11} C_{11} + a_{12} C_{12} + a_{13} C_{13} + a_{14} C_{14} \\ &= (1) (-1)^{1+1} \det \begin{bmatrix} -2 & 0 & 0 \\ -2 & 3 & 0 \\ -2 & 500 & 4 \end{bmatrix} + 0 + 0 + 0 \\ &= (1) (1) \left[(-2) (-1)^{1+1} \det \begin{bmatrix} 3 & 0 \\ 500 & 4 \end{bmatrix} \right] \\ &= (1) \left[(-2) (1) \left[(3)(4) - (0)(500) \right] \right] \\ &= (1) (-2) (3) (4) = -24 \end{aligned}$$



Theorem

If A is a triangular matrix, then $\det(A)$ is the product of the entries on the main diagonal of A



Example

Find $\det(A)$ where $A = \begin{bmatrix} 7 & 4 & 12 & 20 \\ 0 & 5 & 10 & 10 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & -5 \end{bmatrix}$

Upper Triangular

$$\begin{aligned} \det(A) &= (7)(5)(-3)(-5) \\ &= (35)(15) = 525 \end{aligned}$$



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